



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.7

The Distributive Property

Lesson 6-14 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can use the distributive property to expand and factor expressions.

Language Objective: I can explain my steps using the words distributive property, factor, expand, and equivalent.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

The Distributive Property

Distributive Property

Factor

Expand

Equivalent

Coefficient

Like terms



Tap any word to see what it means and a picture.

1

— Multiply a factor by every term inside parentheses.

2

The rule that $a(b + c) = ab + ac$ is the .

3

To rewrite a sum as a product, like $3x + 12 = 3(x + 4)$, is to

4

To multiply out a product over a sum, like $3(x + 4) = 3x + 12$, is to

5

Two expressions that always have the same value are

6 A number multiplied by a variable is a .

7 Terms with the same variable part are .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Use the distributive property to find the total cost. Which way of calculating is easier for you?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **The Distributive Property** — Multiply a factor by every term inside parentheses.
La propiedad distributiva — Multiplicar un factor por cada término dentro de los paréntesis.

☐ **Distributive Property** — Multiplying a number by everything inside the parentheses: $a(b + c) = ab + ac$.
Propiedad distributiva — Multiplicar un número por todo lo que está dentro del paréntesis: $a(b + c) = ab + ac$.

☐ **Factor** — A factor of a whole number divides it evenly, with no remainder. For example, 3 is a factor of 12 because $12 \div 3 = 4$.
Factor — Un factor de un número entero lo divide exactamente, sin dejar residuo. Por ejemplo, 3 es un factor de 12 porque $12 \div 3 = 4$.

☐ **Expand** — To multiply out the parentheses in an expression.
Desarrollar — Multiplicar lo que está dentro del paréntesis en una expresión.

☐ **Equivalent** — Expressions that always have the same value.
Equivalente — Expresiones que siempre tienen el mismo valor.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

$8(12 + 9) = 8 \times \underline{\hspace{1cm}} + 8 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$. I think the form is easier because .

Di tu respuesta primero: Mi respuesta es porque .

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The distributive property multiplies the 3 by each item, giving the same total.

Evidence: Students explain the 3 distributes to both the ticket and the snack ($3 \times 15 + 3 \times 5 = 45 + 15 = 60$), matching 3×20 , and use the word equivalent.

Reasoning: This evidence connects the definition of the distributive property to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- $8(12 + 9) = 8 \times \underline{\quad} + 8 \times \underline{\quad} = \underline{\quad} + \underline{\quad} = \underline{\quad}$. I think the $\underline{\quad}$ form is easier because $\underline{\quad}$.
- My answer is $\underline{\quad}$.

Mi respuesta es $\underline{\quad}$.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is $\underline{\quad}$.
- I know because $\underline{\quad}$.

Mi afirmación es $\underline{\quad}$.

Lo sé porque $\underline{\quad}$.

- So $\underline{\quad}$.

Entonces $\underline{\quad}$.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** The Distributive Property
2. **Notes 2:** Distributive Property
3. **Notes 3:** Factor
4. **Notes 4:** Expand
5. **Notes 5:** Equivalent
6. **Notes 6:** Coefficient
7. **Notes 7:** Like terms
8. **Watch for this mistake:** *The most common mistake in The Distributive Property is multiplying the outside factor by only the FIRST term inside the parentheses and forgetting the second — for example, writing $3(x + 4) = 3x + 4$ instead of $3x + 12$. Before you submit, check: did the outside number reach BOTH terms inside the parentheses?*
9. **Try It 1:** A. $10 + 5n$ — *Distribute the 5 to each term: $5 \times 2 + 5 \times n = 10 + 5n$.*
10. **Try It 2:** A. $4a + 8$ — *Distribute the 4 to each term: $4 \times a + 4 \times 2 = 4a + 8$.*